Multi-Source Candidate Data Transformer – Technical Design

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Problem Statement

Design a deterministic pipeline that converts heterogeneous candidate information from structured (Recruiter CSV, ATS JSON) and unstructured (Resume PDF) sources into a single canonical profile. The system must normalize data, merge conflicting values, preserve provenance, assign confidence scores, and support runtime-configurable output schemas without modifying application code.

Pipeline

Input Sources → Parsing → Extraction → Normalization (phones, skills, dates) → Canonical Candidate Model → Merge & Conflict Resolution → Confidence & Provenance → Runtime Projection → Validation → JSON Output

Canonical Schema & Normalization

Canonical fields: candidate_id, full_name, emails, phones, location, links, headline, years_experience, skills, experience, education, provenance and overall_confidence. Phones are normalized to E.164, dates to YYYY-MM, skills to canonical names (e.g., ReactJS→React), and unknown values remain null rather than fabricated.

Merge & Conflict Resolution

Aspect	Policy
Matching	Merge profiles representing the same candidate.
Priority	Recruiter CSV > ATS JSON > Resume PDF.
Names	Prefer the most complete non-empty value.
Emails/Phones	Normalize and deduplicate.
Skills	Canonicalize then merge unique values.
Experience/Education	Merge unique records.
Confidence	Retain highest confidence while preserving provenance.

Confidence Policy

Recruiter CSV: 0.95 | ATS JSON: 0.90 | Resume PDF: 0.80. These scores reflect source reliability and help downstream systems assess trustworthiness.

Runtime Configurable Projection

A projection layer separates the canonical model from consumer-specific schemas. Configuration supports field selection, renaming, nested mapping, array projection, provenance/confidence toggles, and missing-value handling (null, omit, or error). Projected output is validated before returning.

Edge Cases

• Missing or malformed input files do not terminate processing.
• Conflicting values are resolved using deterministic priority.
• Duplicate skills with different spellings are canonicalized.
• Duplicate emails/phones are removed after normalization.
• Incomplete records preserve known values while leaving unknown fields null.

Scope Limitations

Due to time constraints, LinkedIn/GitHub APIs, OCR for scanned resumes, ML-based entity extraction, fuzzy matching, and distributed processing were intentionally excluded. The modular architecture allows these features to be added later with minimal changes.